Discrepancy Analysis of Conceptual & Concrete Architecture of OpenCV

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SourceArchitects Group Members:

- Sheida Khodabakhshian Khansari
- Vrinda Ashwar
- Matthew Lee
- Werner Bagu-Paiz
- William Moran
- Stefewn Johnson

Top Level Description-Conceptual Architecture

- Layered Architecture
- Core Module [1]
- Imgproc [2]
- Imgcodecs
- Higher-level modules(e.g. features2d)[3]

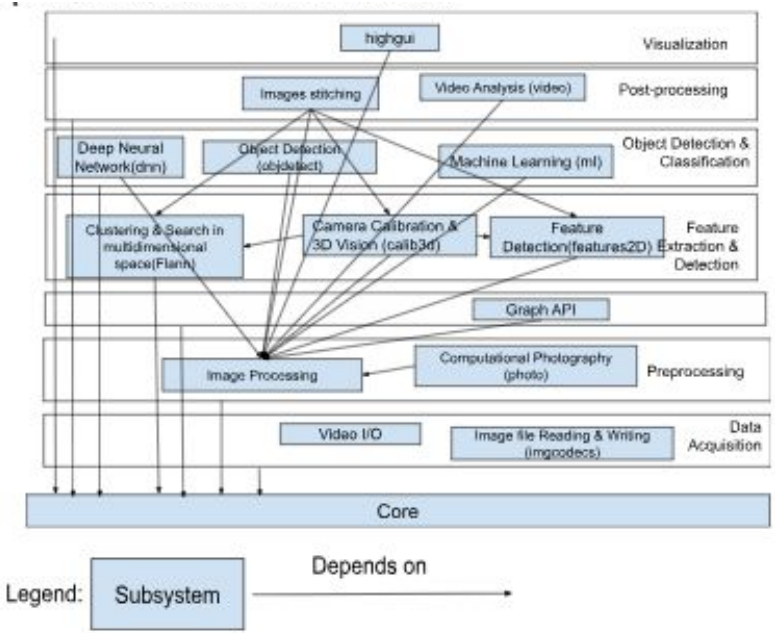

Figure 1. Conceptual Architecture

Top Level Description-Concrete Architecture

- Reverse Engineering
- More connected and less hierarchical
- Unexpected dependencies observed
- Highlights Divergence

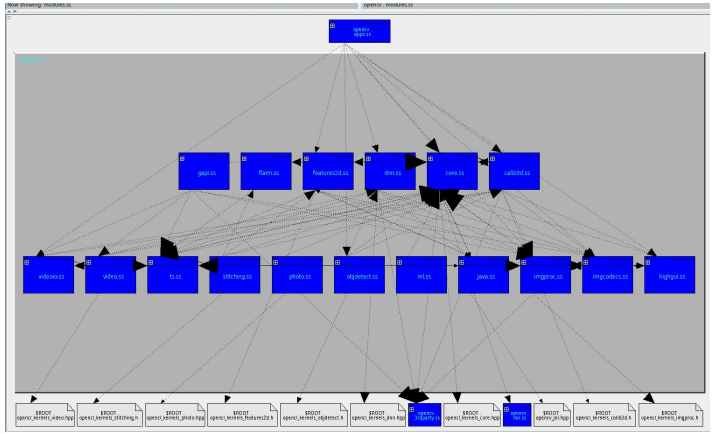

Figure 3: High level architecture of “Module” captured from LSEdit.

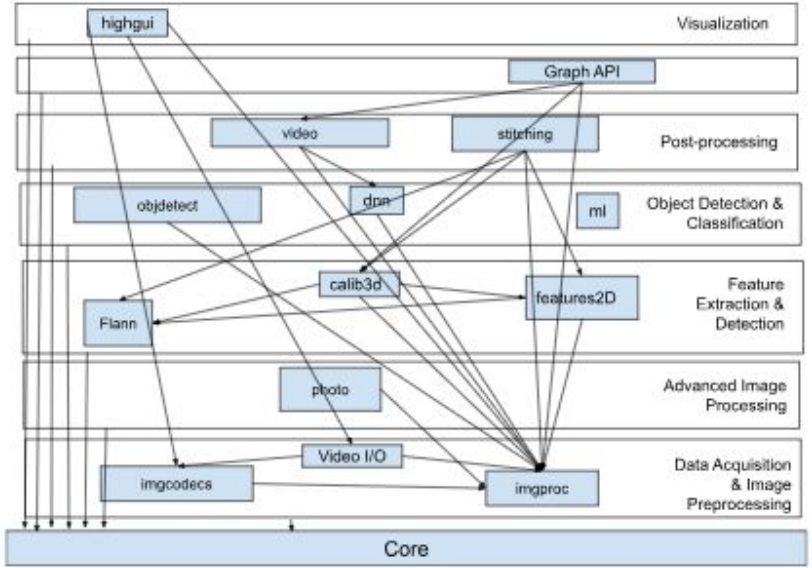

Figure 2. Concrete Architecture

Reflexion Analysis Process

Mapping & Setup

- Analyzed the top level OpenCV architecture
- Examined the imgproc subsystem for deeper relations
- Mapped conceptual subsystem to the concrete one
- Used tools like Understand and LSEdit
- Made sure consistency in subsystem correspondence

Dependency Analysis

- Analyzed the subsystem's interactions and their dependencies
- There are three types of relationships:
  - Divergent
  - Convergent
  - Absent
- Divergent only appears in concrete
- Convergent appears in both concrete and conceptual
- Absent only appears in conceptual
- Graphs visualized the relationship discrepancies very clearly

Rationale & Adjustments

- Investigated for the reasons behind the discrepancies
- Modular refactoring and performance causes were found
- Architectural evolution across the different versions were noted
- Conceptual dependency refinements were proposed
- Pinpointed optional relationship adjustments
- Made sure it resulted in stronger design implementation alignment

Top Level Comparison -Reflexion analysis

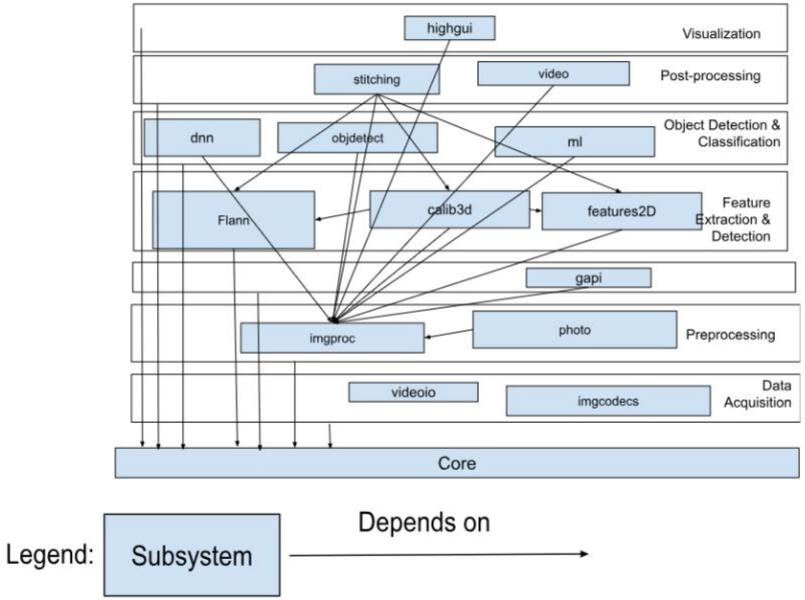

Figure 4. Conceptual Architecture

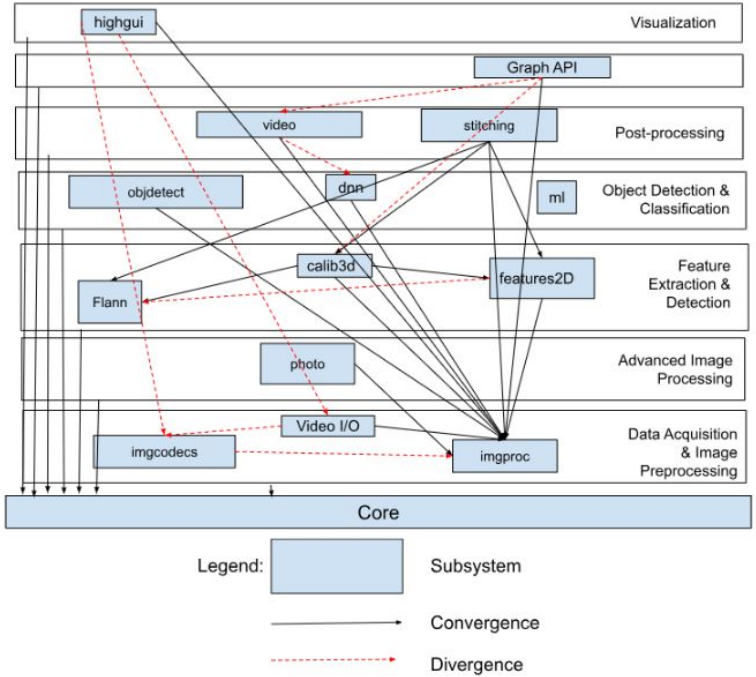

Figure 5. Reflexion Architecture

Rationale for Top Level Differences

Dependency Investigation (Source StickyNotes)

Which?	The module imgcodecs depends on imgproc. Concrete Example From LSEdit: opencv-4.12.0/modules/imgcodecs/src/grfmt_gif.cpp cLinks opencv-4.12.0/modules/imgproc/src/color.cpp
Who?	Super [4]
When?	2024-12-07, part of the initial GIF decoder implementation.
Why?	The imgcodecs module has a required dependency on the imgcodecs module as shown in the build configuration file CMakeLists.txt; “ocv_add_module(imgcodecs opencv_imgproc WRAP java objc python)”. The concrete implementation example of this dependency: ● grfmt_git.cpp depends on on color.cpp as color conversion functionality is used in the GIF decoder. ● GIF decoders use cvtColor() from imgproc to convert decoded images between different color formats [4].

Dependency Investigation (Source StickyNotes)

Which?	The module videoio depends on imgcodecs. Concrete Example From LSEdit: opencv-4.12.0/modules/videoio/src/cap_images.cpp cLinks imgcodecs.ss ->opencv-4.12.0/modules/imgcodecs/src/loadsave.cpp et al.
Who?	Vadim Pisarevsky [5]
When?	2010-05-11
Why?	The videoio module has a required dependency on the imgcodecs module as it needs its functionalities for ● reading/writing individual frames in various image formats ● handling image format conversions during video encoding/decoding ● frame extraction ● video thumbnail generation This dependency is listed in the configuration file : “ocv_add_module(videoio opencv_imgproc opencv_imgcodecs WRAP java objc python)”. The implementation example of this dependency: ● The cap_images.cpp in videoio module implements image sequence capture/writing functionality, which requires the ability to read and write individual image files from disk. ● Calls imread() implemented in loadsave.cpp in imgcodecs module [5].

Dependency Investigation (Source StickyNotes)

Which?	G-API module has an optional dependency on the video module.
Who?	Orest Chura [6]
When?	April 8, 2020
Why?	The dependency of G-API module on video module is optional and and it can function without the video module as show in the CMakeLists.txt: “ocv_add_module(gapi REQUIRED opencv_imgproc OPTIONAL opencv_video opencv_calib3d WRAP python)”. This dependency was added to support video processing operations in G-API’s CPU backend [6]

```
    - opencv_gapi module is linked with opencv_video module (optional dependency)
    - kernels added to a new cv::gapi::video namespace and a brand new files created to provide gapi_video environment
    - there are 2 different kernels as G-API should provide GMat AND GArray<GMat> implementation: cv::calcOptFlowPyrLK doesn't calculate pyramids if vector<Mat> is given so just the cast GMat -> GArray<GMat> wouldn't
represent all the cv:: functionality
    - tests to check both kernels (based on cv::video tests for cv::calcOpticalFlowPyrLK())
    - tests for internal purposes added
    - vectors<T> comparison in tests implemented
    - new (and old too) common test structures refactored to avoid code copypasting
    - "modules/gapi/test/common/gapi_video_tests_common.hpp" created to share some code snippets between perf and acc tests and avoid code copypasting
```

4.x (#16788) · 5.0.0-alpha ··· 4.4.0-openvino

1 parent acede97 commit d50c21e

Filter files...

- modules/gapi
  - CMakeLists.txt
  - include/opencv2/gapi
    - cpu
      - video.hpp
      - imgproc.hpp
      - video.hpp
    - perf
      - common
        - gapi_imgproc_perf_tes...
        - gapi_video_perf_tests.c...

22 files changed +921 -26 lines changed

Search within code

modules/gapi/CMakeLists.txt +8 -1

```
@@ -15,7 +15,12 @@ endif()
 15  15
 16  16  set(the_description "OpenCV G-API Core Module")
 17  17
 18      - ocv_add_module(gapi opencv_imgproc)
     18  + ocv_add_module(gapi
     19  +     REQUIRED
     20  +       opencv_imgproc
     21  +     OPTIONAL
     22  +       opencv_video
     23  + )
 19  24
 20  25  if(MSVC)
```

Figure 6. Code commit that added G-API module’s optional dependency on video module.

Dependency Investigation (Source StickyNotes)

Which?	G-API module has an optional dependency on the video module. Concrete Example From LSEdit: opencv-4.12.0/modules/gapi/test/cpu/gapi_ocv_stateful_kernel_test.cpp cLinks video.ss-> opencv-4.12.0/modules/video/src/bgfg_gaussmix2.cpp
Who?	AsyaPronina (Anastasiya Pronina) [7]
When?	2020-06-04
Why?	The implementation example is a test dependency: ● test file gapi_ocv_stateful_kernel_tests.cpp uses video module functionally(cv::BackgroundSubtractor) to test G-API's stateful kernel functionality. ● Stateful kernel testing framework was introduced in OpenCV 4.4 [7].

Dependency Investigation (Source StickyNotes)

Which?	G-API module has an optional dependency on the calib3d module. Concrete Example From LSEdit: opencv4.12.0/modules/gapi/src/backednds/cpu/gcpustero.cpp cLinks calib3d.ss ->opencv-4.12.0/modules/calib3d/src/stereobm.cpp
Who?	aDanPin [8]
When?	2021-03-31
Why?	The dependency of G-API module on calib3d module is optional and and it can function without the calib3d module as show in the CMakeLists.txt: “ocv_add_module(gapi REQUIRED opencv_imgproc OPTIONAL opencv_video opencv_calib3d WRAP python)”. The concrete implementation example: ● Dependency of the gcpustero.cpp on stereo matching algorithms like StereoBM implemented in calib3d. ● When calib3d is available, G-API can provide stereo vision capabilities and when it is unavailable, the stereo functionality is not included in the build [8].

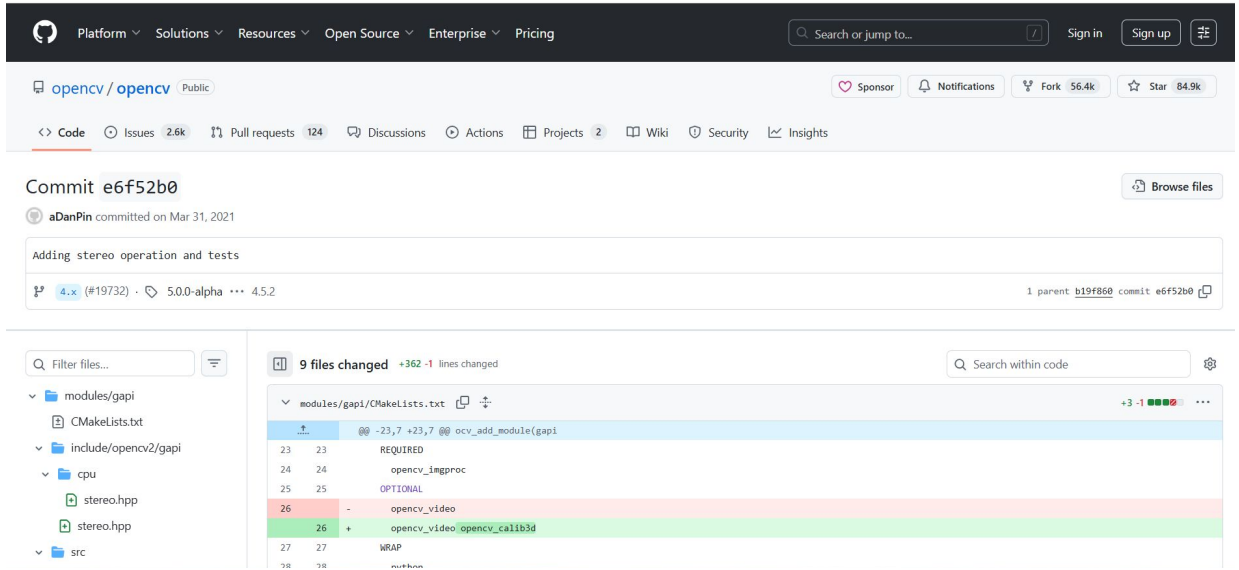

Figure 7. Code commit that added G-API module's optional dependency on calib3d module.

Dependency Investigation (Source StickyNotes)

Which?	Video module has an optional dependency on the dnn module. Concrete Example From LSEdit: opencv-4.12.0/modules/video/src/tracking/tracker_vit.cpp cLinks dnn.ss->opencv-4.12.0 /modules/dnn/src/net.cpp
Who?	lpylpy0514 (PengyuLiu) [9]
When?	2023-09-19
Why?	The dependency of Video module on dnn module is optional and and it can function without the dnn module as show in the CMakeLists.txt: “ocv_add_module(video REQUIRED opencv_core opencv_imgproc OPTIONAL opencv_dnn WRAP python java)”. The video module uses dnn for deep learning-based tracking algorithms. An example is the dependency of the tracker_vit.cpp on dnn functionality (dnn:: Net class implemented in net.cpp) to load and run vision transformer neural network model for object tracking [9].

Dependency Investigation (Source StickyNotes)

Which?	The module features2d has an optional dependency on the flann module. Concrete Example From LSEdit: opencv-4.12.0/modules/features2d/src/matchers.cpp cLinks flann.ss->opencv-4.12.0 /modules/flann/src/miniflann.cpp
Who?	Multiple Contributions: -Vadim Pisarevsky: foundational FLANN wrapper code in miniflann.cpp [10] -Maria Dimashova: created the FlannBasedMatcher class in matchers.cpp [11]
When?	2011-07-13 (Vadim Pisarevsky) [10] 2010-10-29(Maria Dimashova)[11]
Why?	The dependency of features2d module on flann module is optional and and it can function without the video module as show in the CMakeLists.txt “ocv_define_module(features2d opencv_imgproc ${debug_modules} OPTIONAL opencv_flann WRAP java objc python js)”. Example: ● FlannBasedMatcher class in matchers.cpp depends on flann::Index class implemented in miniflann.cpp in flann module for fast approximate nearest neighbor search when matching feature descriptors [10],[11].

Dependency Investigation (Source StickyNotes)

Which?	The module highgui has an optional dependency on imgcodecs. Concrete Example From LSEdit: opencv-4.12.0/modules/highgui/include/opencv2/highgui.hpp cLinks imgcodecs.ss ->opencv-4.12.0/modules/imgcodecs/include/opencv2/imgcodecs.hpp
Who?	Alexander Alekhin [12]
When?	2016-08-25
Why?	The dependency of highgui module on imgcodecs module is optional and and it can function without the imgcodecs module as show in the CMakeLists.txt:"ocv_add_module(highgui opencv_imgproc OPTIONAL opencv_imgcodecs opencv_videoio WRAP python)". ● Part of OpenCV 3.0 modularization effort; original highgui module was split into three separate modules including imgcodecs, videoio, and highgui. This dependency was added to maintain backward compatibility from OpenCV 2.x where image I/O functions were part of the original highgui module. ● Dependency is also included to provide convenient access to image I/O functions (imread, imwrite) when users include the highgui header [12].

Dependency Investigation (Source StickyNotes)

Which?	The module highgui has an optional dependency on videoio. Concrete Example From LSEdit: opencv-4.12.0/modules/highgui/include/opencv2/highgui.hpp cLinks opencv-4.12.0/modules/videoio/include/opencv2/videoio.hpp
Who?	Alexander Alekhin [12]
When?	2016-08-25
Why?	The dependency of highgui module on videoio module is optional and and it can function without the videoio module as show in the CMakeLists.txt:"ocv_add_module(highgui opencv_imgproc OPTIONAL opencv_imgcodecs opencv_videoio WRAP python)". ● The highgui header conditionally includes videoio.hpp to provide convenient access to video I/O functions VideoCapture and VideoWriter when highgui is used. ● This maintains backward compatibility from OpenCV 2.x as these functions were part of the original highgui module[12].

Conceptual Architecture of imgproc Module

- Three major functional layers:
  - Low-Level Pixel Transformations
  - Structural & Geometric Operations
  - High-Level Vision Components
- Clean and modular architecture with minimal cross-layer dependencies
- Follows the standard image-processing pipeline: pixel → geometry → regions/features
- Assumes minimal cross-module interaction and clear hierarchical flow.
- Treats categories like color operations, morphology, filtering, and segmentation as isolated functional blocks, not intertwined implementations.

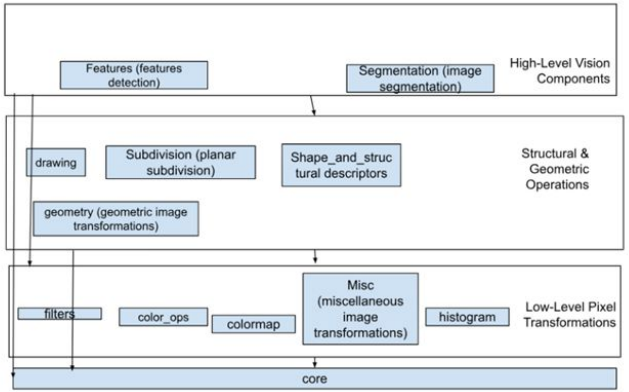

Figure 8. Conceptual Architecture of imgproc module.

Concrete Architecture of imgproc

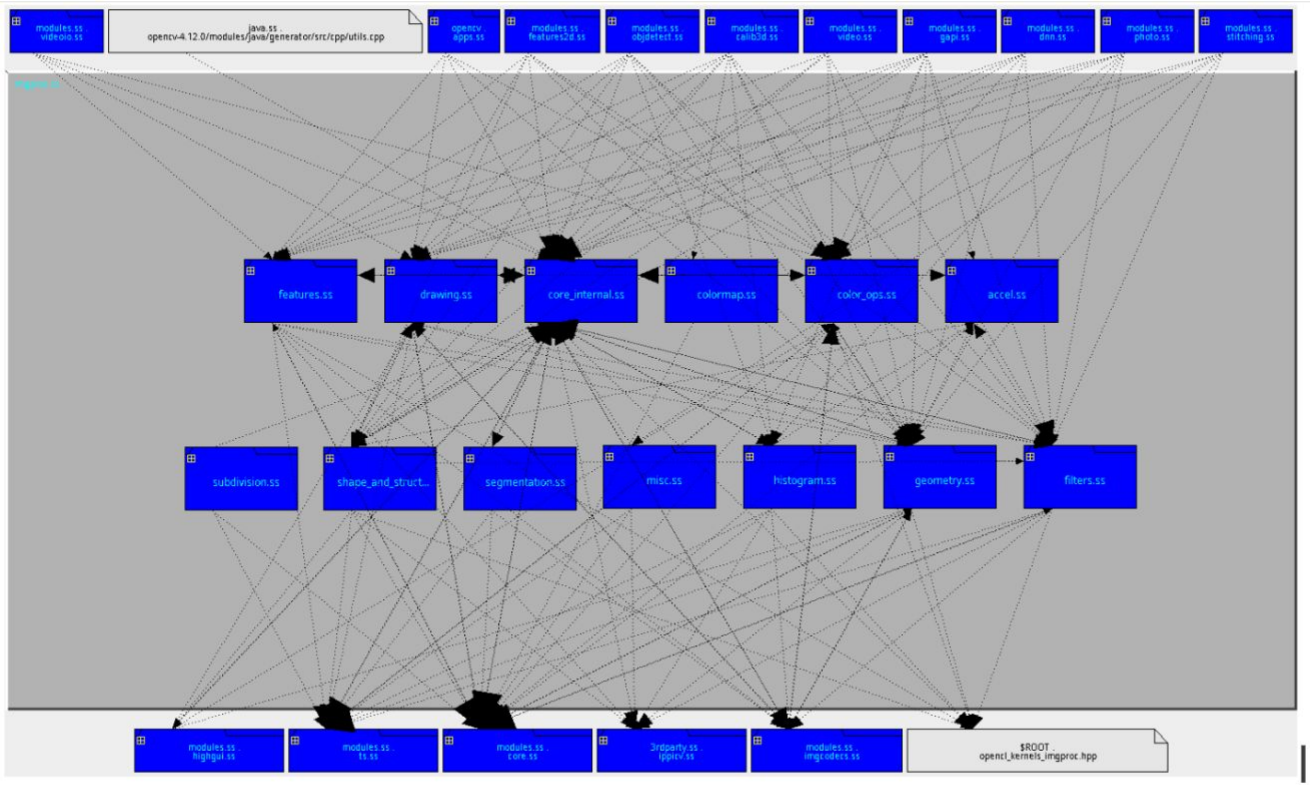

Figure 9: Concrete architecture of imgproc captured from LSEdit using the listed subsystems.

Concrete Architecture of imgproc Module

- Concrete architecture is not layered, instead, it is a tightly interconnected set of subsystems.
- Many modules act as shared utilities, especially filters.ss and core_internal.ss, creating high fan-in and fan-out.
- Some conceptual categories (e.g., object detection and motion analysis) exist only as documentation tags, not real implementations.
- Reveals a structure driven by performance, reuse, and hardware optimization, not by conceptual hierarchy.

Key subsystems:

- filters.ss
- geometry.ss
- morphology.ss
- segmentation.ss
- features.ss
- core_internal.ss
- color_ops.ss

Subsystem Level Comparison - Reflexion Analysis

Missing Modules in imgproc

- “Object detection” and “motion analysis/object tracking” appear in OpenCV’s imgproc documentation
- No actual implementations exist in the imgproc codebase
- These modules do not appear in the LSEdit concrete architecture
- @defgroup imgproc_object and @defgroup imgproc_motion are documentation placeholders only

Reflexion Analysis Insight

- Conceptual model assumes clean, separated functional layers
- Concrete architecture shows utility sharing, internal coupling, and performance driven structure
- Conceptual view overestimates modular separation
- Real implementation shaped by hardware optimizations (SIMD, SSE, AVX)

Rationale for Subsystem Level Differences

- Performance Improvements
  - Many imgproc subsystems reuse the same optimized routines (filtering, interpolation, matrix operations)
- Math and Computational Dependencies
  - Geometry, segmentation, features and color operations all depend heavily on the same Core utilities
- Hardware Acceleration
  - SIMD/SSE/AVX are placed in low-level files increasing cross-links
- Documentation categories vs. real modules
  - “Object detection” and “motion analysis/object tracking” exist only as documentation groups

How We Determined These Rationales

- LSEdit Dependency Graphs
  - Visualized cross-module links and subsystem coupling
  - Identified unexpected dependencies (e.g. geometry.ss → filters.ss)
- Understand Dependency Traces
  - Verified which subsystems reused shared utilities
- Source Code Inspection
  - Examined imgproc and found that many operations are implemented across multiple files
- Verification of Missing Modules
  - Checked for “object detection” and “motion analysis/object tracking” folders/files

Concurrency and Team Issues

- Concurrency Effects
  - Performance requirements introduce cross-module dependencies
  - Modules rely heavily on Core for parallelization
  - Concurrency optimizations cause modules to bypass conceptual layering
  - HAL adds additional implicit dependencies
- Team Issues
  - Multiple teams maintain different modules
  - Inconsistent design decisions lead to unplanned couplings
  - Backward compatibility prevents refactoring increasing architectural drift
  - Concrete model is more interconnected compared to Conceptual model

Diagrams

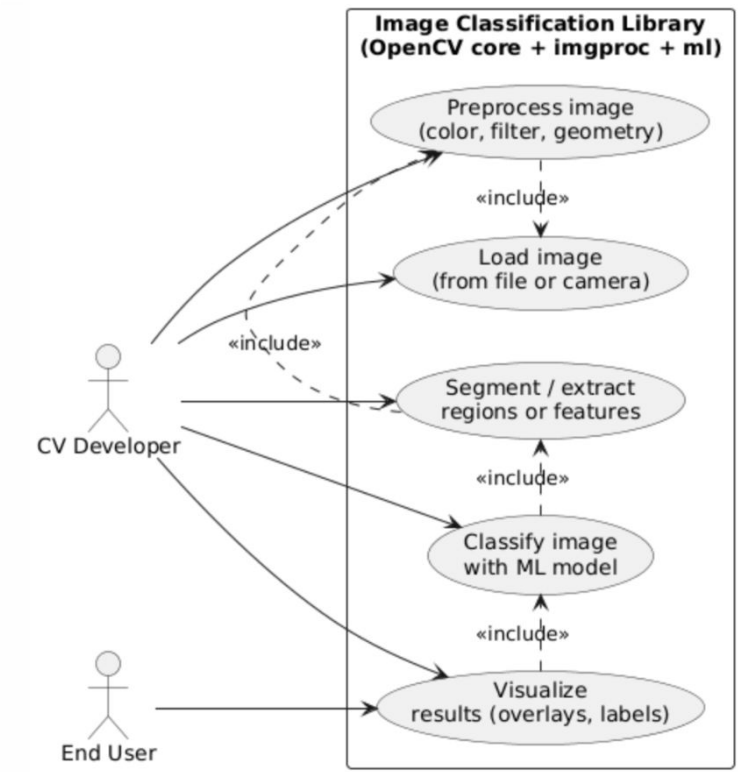

Figure 10 . Use case diagram

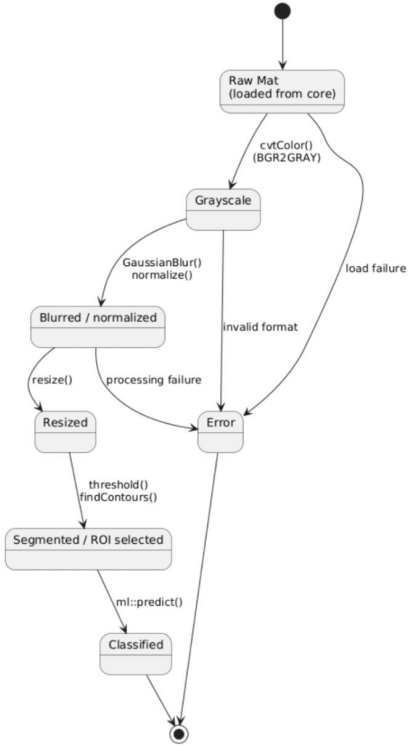

Figure 11 . State diagram

Diagrams

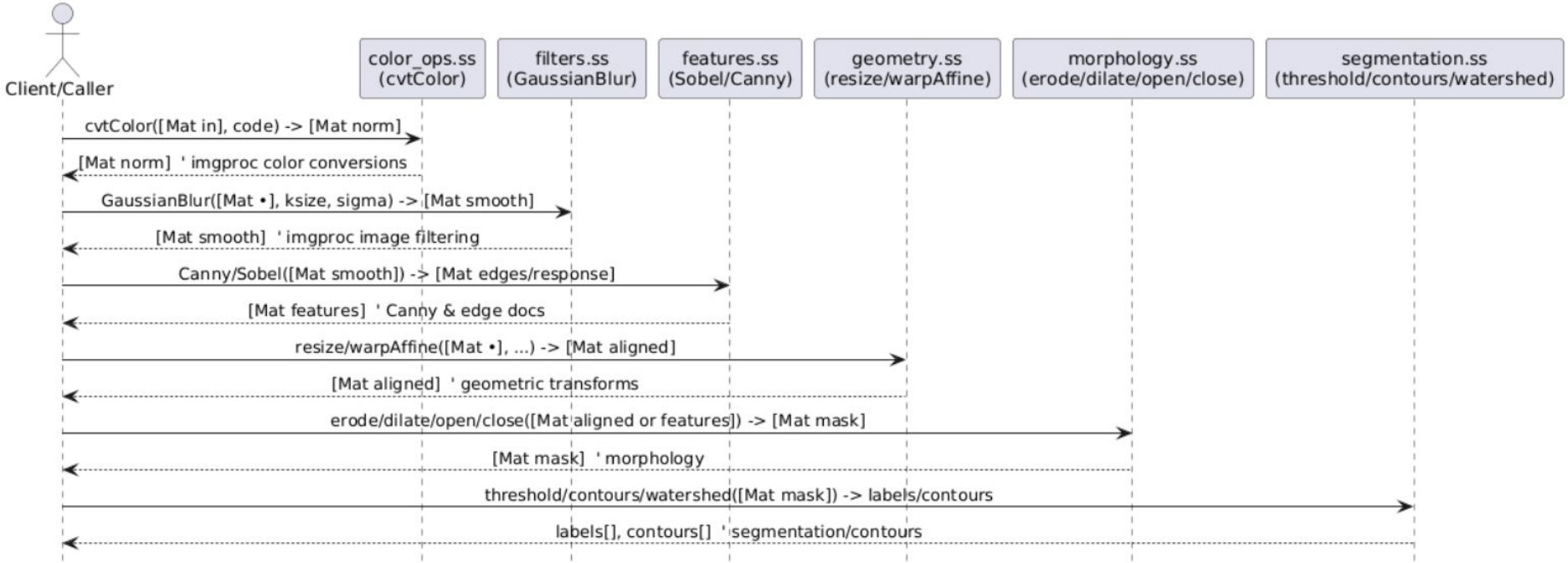

Figure 12 [13]. Sequence Diagram of Data and Control Flow Across imgproc Subsystems

Limitations Of Reported Findings

- Dependencies were missing sometimes from Understand and LSEdit
- Optional modules would cause complications in dependency interpretation during analysis
- Conceptual architecture underestimated the actual module coupling
- Hidden cross-module connections were introduced when doing hardware optimizations
- Object detection modules were the only placeholders
- Analysis was restricted to only static tool observations

Learned Lessons

- Reflexion analysis bought clarity for the conceptual against concrete architecture
- Tools like Understand and LSEdit discovered hidden subsystem interactions
- Comparing at the highest level identified unexpected architectural divergences
- Analysis on subsystem showed deeper module coupling than what we expected
- Visual dependency graphs increased our clarity of interactions
- Our findings highlighted iterative refinement for consistency in architecture

References

[1] “Core functionality (core module),” OpenCV Documentation, accessed Oct. 28, 2025. [Online]. Available: https://docs.opencv.org/4.x/d0/de1/group__core.html

[2] “OpenCV: Image Processing.” Accessed: Oct. 06, 2025. [Online]. Available: https://docs.opencv.org/4.x/d7/dbd/group__imgproc.html

[3] “modified features2d interface; added algorithmic test for DescriptorM… · opencv/opencv@69e329c · GitHub.” Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/69e329c9

[4] opencv, “Merge pull request #25691 from redhecker:gifSupport · opencv/opencv@082cd7a,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/082cd7a74eab9a435c414e4fe7a5c0803d4d3b4d

[5] opencv, “‘atomic bomb’ commit. Reorganized OpenCV directory structure · opencv/opencv@127d664,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/127d6649a1c83397bf42799ac807af41aa507b30

[6] “gapi: Full calcOpticalFlowPyrLK implementation (2 overloads) and tests · opencv/opencv@d50c21e · GitHub.” Accessed: Nov. 16, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/d50c21e5

[7] opencv, “Enable stateful kernels in G-API OCV Backend · opencv/opencv@b083c20,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/b083c20eb2eba7227f66dd10837609db9704fec6

[8] opencv, “Adding stereo operation and tests · opencv/opencv@e6f52b0,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/e6f52b0996d8e4f96ed13e501b4166805f09024b

[9] opencv, “Merge pull request #24201 from lpylpy0514:4.x · opencv/opencv@70d7e83,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/70d7e83dca7824e7e9d09e03630d1134628dc70a

References P2

[10] opencv, “upgraded to FLANN 1.6. Added miniflann interface, which is now used i… · opencv/opencv@562914e,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/562914e33bea921a0a235fbf2b6bba6830644e16

[11] “modified features2d interface; added algorithmic test for DescriptorM… · opencv/opencv@69e329c · GitHub.” Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/69e329c9

[12] opencv, “fix modules deps · opencv/opencv@063ac74,” GitHub. Accessed: Oct. 28, 2025. [Online]. Available: https://github.com/opencv/opencv/commit/063ac74369199c1cfcaefe266d963ea1c21db75d

[13] “PlantUML,” PlantUML.com, https://plantuml.com/ (accessed Oct. 28, 2025).